PS6: Build a prototype AI system that detects: Safety gear compliance

Factories face constant risk from unsafe worker practices and hazardous incidents such as smoke and fire. Yet detection and response are often delayed or inconsistent. Many industrial sites rely on manual monitoring, periodic inspections, or standalone alarms that may not quickly identify unsafe gear usage or provide timely, location-specific evidence of smoke or fire. This leads to slower evacuation, increased damage, and higher risk of injury or worse outcomes.

There is a need for an AI-based, real-time monitoring system that can automatically detect whether workers are wearing required safety gear and identify early signs of smoke and fire within factory premises. The solution should operate reliably in dynamic industrial environments, support fast alerting to the right personnel, and help reduce incident response time by providing actionable insights such as the affected area and detected event type.

Hackathon Goal:

Build a prototype AI system that detects:

Safety gear compliance (e.g., helmets, high-visibility vests, protective footwear, gloves where applicable), Smoke presence, Fire presence in real time from CCTV/video feeds or recorded footage, and generates alerts with location/context to enable faster intervention and improved safety compliance across factory operations.